Chapter 3

Task C: Computational Exercise

3.1 The Dynamic Programming Problem

In this problem, higher current production by the government increases immediate utility. However, it simultaneously poses a risk to future production by polluting the air. Consequently, the government must address this trade-off through a dynamic optimization approach.

We consider an agent who chooses how much of the current stock S_t to use or invest, denoted Y_t, to maximize expected discounted utility over an infinite horizon. The problem is affected by a shock δ_t, which follows a two-state Markov chain with possible values $\{1, 0.8\}$ and transition probabilities η (probability of switching states) and $1-\eta$ (probability of remaining in the same state). The growth parameter is $\tau = 0.01$.

The recursive form of the Bellman equation is:

$$V(S_t, \delta_t) = \max_{Y_t > 0} \Big\{ U(Y_t) + (1-\beta)\, \mathbb{E}\big[V(S_{t+1}, \delta_{t+1}) \mid \delta_t \big] \Big\},$$

subject to the law of motion for the state variable:

$$S_{t+1} = (1+\tau) S_t - \delta_t Y_t,$$

and the constraints:

$$Y_t > 0, \text{ or } \quad \frac{(1+\tau) S_t - S_{t+1}}{\delta_t} > 0.$$

Alternatively, by defining $Y_t = \frac{(1+\tau)S_t - S_{t+1}}{\delta_t}$, we can rewrite the Bellman equation using S_{t+1} as

the control variable:

$$V(S_t, \delta_t) = \max_{S_{t+1}} \Big\{ U\Big(\frac{(1+\tau)S_t - S_{t+1}}{\delta_t}\Big) + (1-\beta)\Big[(1-\eta)V(S_{t+1}, \delta_t) + \eta V(S_{t+1}, \bar{\delta}_t)\Big]\Big\},$$

subject to

$$\frac{(1+\tau)S_t - S_{t+1}}{\delta_t} > 0 \quad \text{or} \quad S_{t+1} < (1+\tau)S_t.$$

State and Choice Variables

State Variables: S_t (current stock) and δ_t (shock). **Choice Variable:** Y_t (usage/investment) or equivalently S_{t+1} (next-period stock).

State variables represent information inherited from the past, while choice variables are decisions made in the current period.

3.2 First-Order Conditions and Envelope Theorem

The per-period objective function is:

$$F(S_t, S_{t+1}, \delta_t) = U\Big(\frac{(1+\tau)S_t - S_{t+1}}{\delta_t}\Big) + (1-\beta)\Big[(1-\eta)V(S_{t+1}, \delta_t) + \eta V(S_{t+1}, \bar{\delta}_t)\Big].$$

The Bellman equation is:

$$V(S_t, \delta_t) = \max_{S_{t+1}} F(S_t, S_{t+1}, \delta_t).$$

3.2.1 First-Order Condition (FOC)

Differentiating F with respect to S_{t+1} and setting it to zero, we get:

$$-\frac{1}{\delta_t} U'\Big(\frac{(1+\tau)S_t - S_{t+1}}{\delta_t}\Big) + (1-\beta)\Big[(1-\eta)V_S(S_{t+1}, \delta_t) + \eta V_S(S_{t+1}, \bar{\delta}_t)\Big] = 0.$$

Rearranging gives:

$$\frac{1}{\delta_t} U'\Big(\frac{(1+\tau)S_t - S_{t+1}}{\delta_t}\Big) = (1-\beta)\Big[(1-\eta)V_S(S_{t+1}, \delta_t) + \eta V_S(S_{t+1}, \bar{\delta}_t)\Big].$$

3.2.2 Envelope Theorem

Using the envelope theorem, the derivative of V with respect to S_t is:

$$V_S(S_t, \delta_t) = \frac{(1+\tau)}{\delta_t} U'\Big(\frac{(1+\tau)S_t - S_{t+1}}{\delta_t}\Big).$$

3.3 Euler Equation and Transversality Condition

Euler Equation

Combining the FOC and the envelope condition, the Euler equation is:

$$\frac{1}{\delta_t} U'\big(c_t\big) = (1-\beta)\Big[(1-\eta)\frac{(1+\tau)}{\delta_t} U'\big(c_{t+1}^{(\delta_t)}\big) + \eta\frac{(1+\tau)}{\bar{\delta}_t} U'\big(c_{t+1}^{(\bar{\delta}_t)}\big)\Big],$$

where

$$c_t = \frac{(1+\tau)S_t - S_{t+1}}{\delta_t}, \quad c_{t+1}^{(\delta_t)} = \frac{(1+\tau)S_{t+1} - S_{t+2}}{\delta_t}, \quad c_{t+1}^{(\bar{\delta}_t)} = \frac{(1+\tau)S_{t+1} - S_{t+2}}{\bar{\delta}_t}.$$

Transversality Condition

The transversality condition ensures no unbounded accumulation of S_t:

$$\lim_{t\to\infty} (1-\beta)^t V_S(S_t, \delta_t) S_t = 0.$$

This prevents explosive solutions and ensures optimal finite paths for $\{S_t\}$.

Interpretation

State vs. Control: State variables (S_t, δ_t) summarize inherited information, while the choice variable (S_{t+1}) determines allocation between current and future periods. **FOC and Envelope:** The FOC equates marginal cost and marginal benefit of adjusting S_{t+1}, while the envelope condition captures the sensitivity of value to S_t. **Euler Equation:** Links current marginal utility to expected discounted future marginal utility. **Transversality:** Ensures no unbounded accumulation or pathological solutions.

3.4 Value Function Iteration Method

The Value Function Iteration (VFI) method is a numerical approach used to solve dynamic programming problems. The main steps are as follows:

1. Discretize the state space S_t and the shock space δ_t.

2. Initialize the value function $V(S_t, \delta_t)$, often with an initial guess.
3. Iterate over the Bellman equation:

$$V^{(n+1)}(S_t, \delta_t) = \max_{S_{t+1}} \left\{ U\left(\frac{(1+\tau)S_t - S_{t+1}}{\delta_t} \right) + (1-\beta)\mathbb{E}[V^{(n)}(S_{t+1}, \delta_{t+1})] \right\}.$$

4. Stop iterating when the value function converges, i.e., $\|V^{(n+1)} - V^{(n)}\| < \epsilon$ for a small tolerance ϵ.

For further details on the implementation method, refer to the Julia code implementation report.

Graphs of Value and Policy Functions

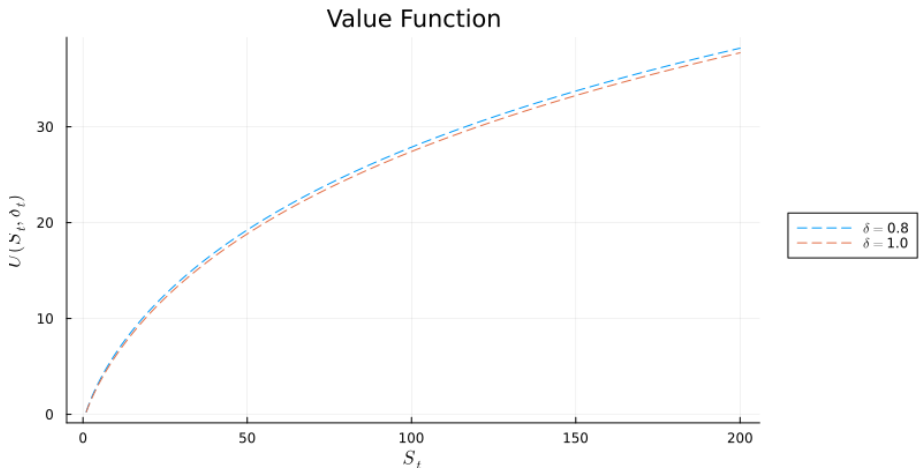


Figure 3.1: Value Function as a Function of Remaining Stock S_t

Interpretation of the Graphs

Figure 3.1 shows the value function $V(S_t, \delta_t)$ for different levels of the stock S_t under the two shock states ($\delta_t = 1$ and $\delta_t = 0.8$). The value function is higher when $\delta_t = .08$ than when $\delta_t = 1$. This is because if the air is less susceptible to production, the healthy air reservoir is less likely to be destroyed.

Figure 3.2 depicts the policy function, showing the optimal usage Y_t as a function of S_t for the two shock states. When $\delta_t = 0.8$, the government produces more (higher Y_t) compared to when $\delta_t = 1$. A lower δ means less damage to the clean air stock per unit of production. Hence, production increases as its relative price decreases.

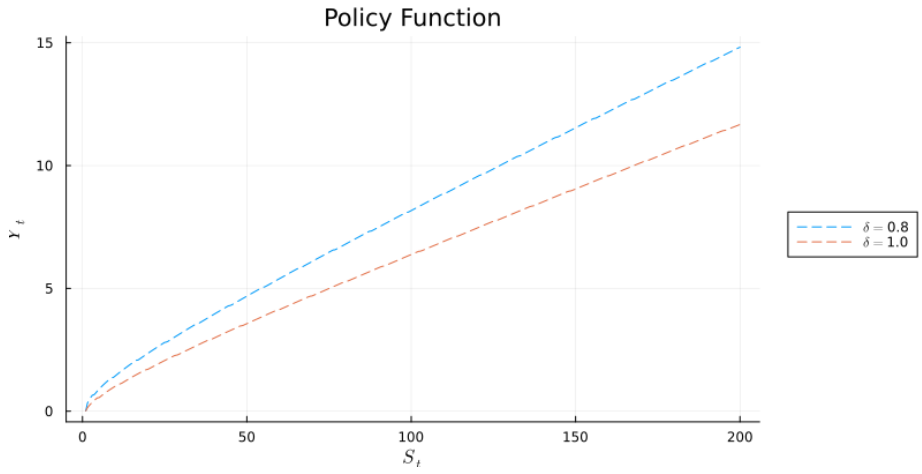


Figure 3.2: Policy Function for Optimal Usage Y_t as a Function of S_t

Effect of Initial δ_t

The initial shock δ_t has a significant impact on both the value and policy functions:

- A higher δ_t increases the value function, reflecting greater utility potential from the stock.
- The policy function exhibits higher optimal usage levels Y_t for higher δ_t, as the stock is more productive when δ_t is favorable.

This dynamic highlights the importance of the shock's state in shaping optimal decision-making in the model.

3.5 Simulation Results and Interpretation

Interpretation of the Simulation

1. Random Behavior of δ_t: The first panel in Figure 3.3 shows the evolution of δ_t, representing the marginal negative effect of production (Y_t) on the stock of clean air (S_t). The stochastic nature of δ_t alternates between high ($\delta_t = 1$) and low ($\delta_t = 0.8$) states. Higher δ_t reflects greater environmental damage, while lower δ_t indicates reduced impact.

2. Decline in Clean Air Stock (S_t): The second panel illustrates a consistent decline in the stock of clean air (S_t) over time, driven by the depletion effect of production activities (Y_t). During periods of high δ_t, the decline in S_t is more pronounced, reflecting greater environmental degradation. Conversely, periods of low δ_t exhibit slower depletion.

3. Government Production Decisions (Y_t): The third panel depicts the government's output (Y_t). When δ_t is high, the government reduces production to limit environmental damage and preserve clean air. In contrast, during low δ_t periods, production increases as the marginal

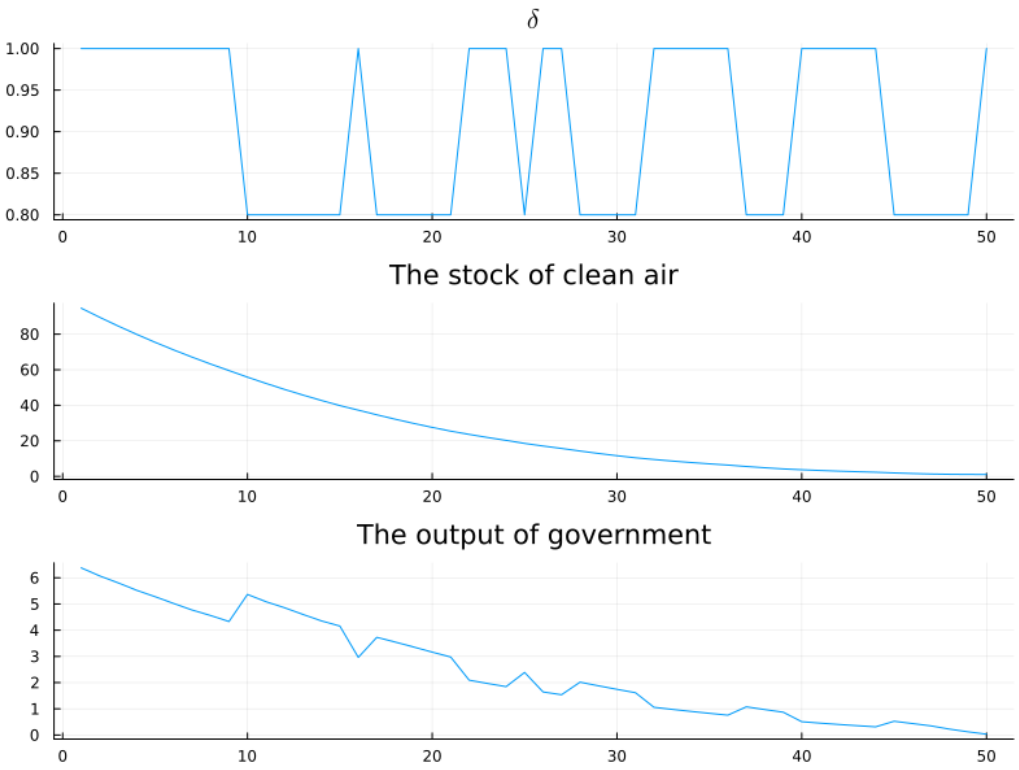


Figure 3.3: Simulation of δ_t, Stock of Clean Air (S_t), and Government Output (Y_t)

damage is less severe.

4. Transitions Between States: During transitions from high to low δ_t (or vice versa), the government adjusts production accordingly. For example, production increases when δ_t decreases, reflecting the lower cost of pollution. Conversely, when δ_t rises, the government reduces output to mitigate environmental harm.

5. Long-Term Sustainability: The simulation highlights a trade-off between production and environmental preservation. Over time, S_t declines, emphasizing the need for careful balancing of current output and future environmental sustainability. The model illustrates the importance of responsive policies to manage stochastic shocks effectively.

3.6 Further Analysis (Optional)

Adjusting the parameters reveals the following: increasing τ (the regeneration parameter of clean air) results in a rise in production; however, the overall trend remains downward. Similarly, decreasing β (the probability of government dismissal) leads to an increase in the future value and delays the decline in production, though the downward trend persists.

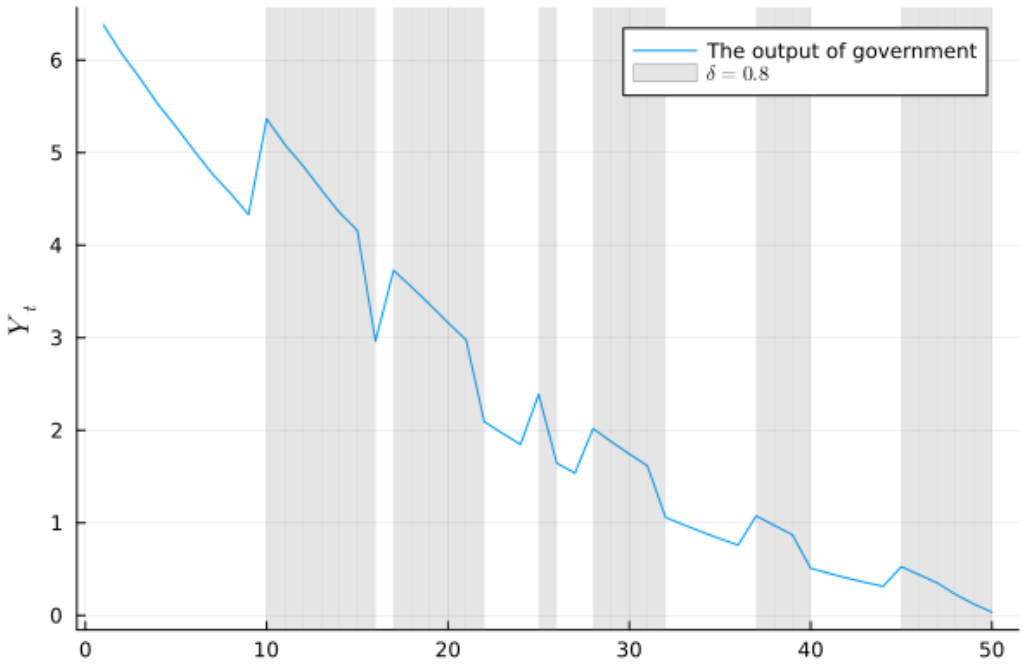


Figure 3.4: Government Output (Y_t) and δ_t